Essays

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Give an account on the structure of proteins and their roles in the cell membrane.

(10 marks)

##

Structure of proteins (Max. 3)

- Amino acids are linked by peptide bonds to form a polypeptide chain. (1)
- The polypeptide chain coils / folds up to form a helical or parallel structure. (1)
- The polypeptide chain may further coil and fold to form a three-dimensional structure. (1)
- When more than one polypeptide chains bind together, they form a protein molecule with specific conformation. (1)

Roles of proteins in the cell membrane (Max. 4)

- Proteins stabilize the membrane structure. (1)
- Channel proteins allow ions and water soluble substances to pass through. (1)
- Carrier proteins are involved in active transport. (1)
- Membrane-bound enzymes catalyse chemical reactions. (1)
- Glycoproteins may act as recognition markers. (1)
- Receptor proteins bind with specific hormones / messengers which stimulate or inhibit certain cellular activities. (1)

Communication (Max. 3)

(10 marks)

##

Skills assessed	Marks
Understanding basic concepts	7
Communication	3

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Discuss osmosis and its importance to organisms. (12 marks)

##

Definition of osmosis

- Osmosis is the net movement of water molecules from a region of higher water potential to a region of lower water potential across a differentially permeable membrane (1)

Effects of osmosis to cells

- When plant cells are bathed in a hypotonic solution, the cells gain water and become turgid. (1)
- When plant cells are bathed in a hypertonic solution, the cells lose water and become plasmolysed. (1)
- When animal cells are bathed in a hypotonic solution, the cells gain water and lyse (e.g. haemolysis). (1)
- When animal cells are bathed in a hypertonic solution, the cells lose water and shrink. (1)

Importance to organisms (Max. 4)

- Animals:
  - Water in the lumen of the intestines is absorbed by osmosis. (1)
  - Water in the glomerular filtrate is reabsorbed by osmosis. By adjusting the permeability of the kidney tubules to water, the amount of water excreted in urine can be controlled. (1)
- Plants:
  - Water in the soil is absorbed into the roots by osmosis. (1)
  - Plant cells gain water by osmosis and become turgid. Turgidity gives support to plants. (1)
- Daily lives:
  - Humans make use of the principle of osmosis to preserve food by soaking food in concentrated sugar or salt solution. (1)
  - Humans make use of the principle of osmosis to design dialysis machines for haemodialysis for patients with kidney failure. (1)

Communication (Max. 3)

(12 marks)

##

Skills assessed	Marks
Understanding basic concepts	5
Applying concepts	4
Communication	3

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|!|BCEEB0040000003|!| ★

Biological membranes are found enclosing cells and membrane-bound organelles.

Discuss the functions of these membranes. (11 marks)

##

Membrane-bound structures	Functions of membranes
Cell	• gives shape to the cell • provides mechanical support to the cell • prevents leakage of the cell contents • separates the cell from the surrounding medium • contains receptor proteins that can recognize specific chemical substances and other cells • controls the movement of substances into and out of cell
Nucleus	• gives shape to the nucleus • separates genetic materials from the cytoplasm
Mitochondrion	• outer membrane gives the mitochondrion its shape • inner membrane contains enzymes for ATP synthesis / respiration
Chloroplast	• gives the chloroplast its shape • internal sac-like membranes contain chlorophyll for carrying out photosynthesis
Vacuole	• holds the stored nutrients, minerals and metabolic wastes • holds water and gives turgidity to the cell
Endoplasmic reticulum	• with protein-synthesizing ribosomes attached on its membrane • helps in transporting proteins in rough ER • helps in transporting carbohydrates and lipids in smooth ER

(Max. 3)

(Max. 5)

Communication (Max. 3)

(11 marks)

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Skills assessed	Marks
Applying concepts	8
Communication	3

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